# Retail purchasing trends, 2020–2021

Simulated retail analysis · Simona Topliceanu · Excel Firedrill

**Fewer items per order in 2021, but higher spending per item. Average order value rose 3.5%.**

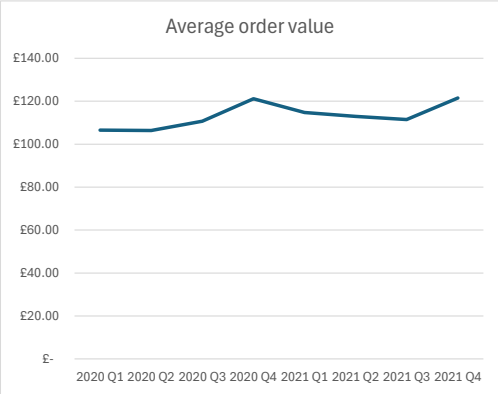
| Annual comparison | 2020 | 2021 | Change |
|---|---|---|---|
| Average order value (AOV) | £111.74 | £115.60 | +3.5% |
| Units per transaction (UPT) | 2.57 | 2.04 | -20.7% |
| Average unit price (AUP) | £43.47 | £56.73 | +30.5% |

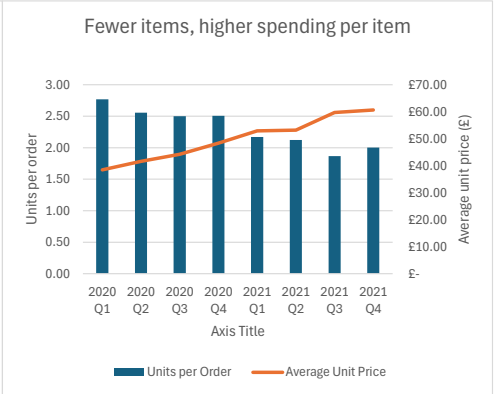
### Quarterly results

AOV = revenue ÷ orders. UPT = units ÷ orders. AUP = revenue ÷ units. Orders are unique order IDs.

| Period | Orders | Units | Revenue | AOV | UPT | AUP | Check |
|---|---|---|---|---|---|---|---|
| 2020 Q1 | 380 | 1,052 | £40,471.50 | £106.50 | 2.77 | £38.47 | OK |
| 2020 Q2 | 520 | 1,329 | £55,314.50 | £106.37 | 2.56 | £41.62 | OK |
| 2020 Q3 | 440 | 1,100 | £48,692.00 | £110.66 | 2.50 | £44.27 | OK |
| 2020 Q4 | 560 | 1,403 | £67,828.50 | £121.12 | 2.51 | £48.35 | OK |
| 2021 Q1 | 430 | 933 | £49,349.00 | £114.77 | 2.17 | £52.89 | OK |
| 2021 Q2 | 470 | 998 | £53,071.50 | £112.92 | 2.12 | £53.18 | OK |
| 2021 Q3 | 450 | 840 | £50,161.50 | £111.47 | 1.87 | £59.72 | OK |
| 2021 Q4 | 590 | 1,182 | £71,685.50 | £121.50 | 2.00 | £60.65 | OK |

### Quarterly trends




**Why it matters: rising spend per item offset a fall in the number of items bought per order.**

### Scope and cleaning

Removed 137 duplicate orders. Excluded 22 orders from 2022 and their 38 item rows. Delivery fees excluded. Standardised dates and prices; trimmed order IDs. Annual KPIs use annual totals. Detailed cleaning log below.

Next question: did customers choose more expensive products, or did prices rise for the same products?

### Cleaning log

| Finding | Action |
|---|---|
| 3,999 original order rows; 137 duplicates | Deduplicated by order ID: 3,862 unique orders. |
| 22 orders dated in 2022 | Excluded: 3,840 orders retained for 2020–2021. |
| 38 item rows linked to 2022 orders | Excluded: 7,576 item rows retained. |
| Three source date formats | Converted to Excel dates. |
| Prices with £, GBP or plain numbers | Converted to numeric unit prices. |
| 258 item-table order IDs with surrounding spaces | Trimmed order IDs before lookup. |
| Delivery fees present | Excluded from item revenue. |